

CSE 4392/5320: SOFTWARE ENGINEERING WITH GENERATIVE AI

Team Project 1, 13%

Due: 09/27/2026 11:59PM submit to Canvas

1 Introduction

This team project requires each team to apply the evolutionary prompt optimization process, shown in Figure 1, to optimize the prompt and the methodology for identifying domain-specific phrases and classifying the phrases to classes, attributes of the classes and relationships between the classes. Each team is required to work with a large language models (LLM) of the teams' choice to perform the activities of the process. The optimization goal is to reach 97% for the F1 score.

2 What Needs to be Done?

1. Use the library, car rental and National Trade Show Service (NTSS) in this order to optimize the identification activity and classification activity, respectively. That is, perform the optimization process in Figure 1 for the identification activity, and then repeat the process for the classification activity. The following describes how to perform the process.
2. Set the optimization goal to F1=97%.
3. Modify the identification/classification methodology so that it is suitable to work with a LLM as well as maximizing the F1 score. Produce an initial prompt for the identification/classification activity. Suggestion: produce the initial methodology and initial prompt manually, then modify them in subsequent iterations either manually or working with a LLM.
4. Prompt the LLM to perform the identification/classification activity using the methodology on the library description.
5. Prompt the LLM to compute the F1 score.
6. If $F1 < 97\%$, then go to step 3, else continue.
7. Prompt the LLM to perform the identification/classification activity using the methodology on the car rental description and the NTSS description, respectively. Compute the F1 scores for these two examples.
8. If one of the F1 scores is less than 97%, then go to step 3, else continue.
9. Apply the Chain of Thought (CoT) prompting technique to integrate the identification and classification activities. Test your CoT prompt on all of the business descriptions. Save the LLM responses and the F1 scores.

Chain of Thought prompting guides a LLM to solve a problem step by step. For example, to generate a domain model, one must identify the domain-specific phrases, then classify the phrases to classes (including association classes), attributes and relationships (inheritance, aggregation and association). Applying Chain-of-Thought prompting, you write a prompt that tells the LLM how to perform the identification step followed by how to perform the classification step.
10. Produce a PPT presentation to share your experience and your project work including the methodologies, optimized prompts, LLM responses and the F1 scores, as well as findings and lessons learned.

The classification result must also include association classes. An association class is needed, and can be found from an association relationship between two classes if both of the following conditions are met:

- The association relationship is a many-to-many relationship such as Student enroll-in Course (a student may enroll in more than one course, and each course may have one or more students), and
- Information about the association (not either of the two classes) needs to be maintained (e.g., we need to record the grade for a student enrolling in a course). If we do not need to record the grade, then we do not need an association class (e.g., we may not need to record grades for training courses).

3 What to Submit

Submit the following:

- The optimized identification methodology and prompt.
- The LLM responses for the identification activity for the library, car rental and NTSS descriptions and their F1 scores.
- The optimized classification methodology and prompt.
- The LLM responses for the classification activity for the library, car rental and NTSS descriptions and their F1 scores.
- The CoT prompt and the LLM responses as well as the F1 scores for the library, car rental and NTSS descriptions.
- The PPT presentation.

4 How to Submit

Submit on Canvas as instructed by the GTA and before the deadline.

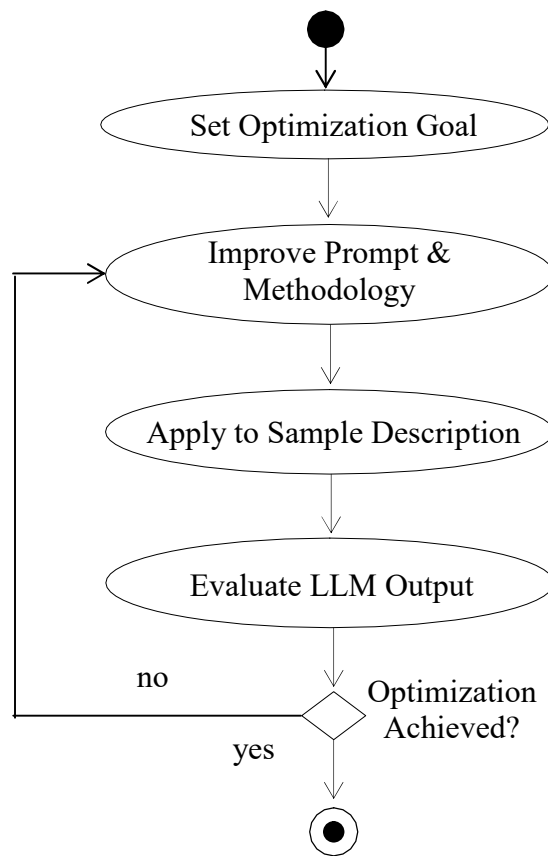


Figure 1: